

Flashcards for Studying: Introduction to Operating Systems (OS)

Card 1: Definition of OS

Q: What is an Operating System (OS)?

A: System software that manages hardware and software resources, acting as an intermediary between users and hardware for efficient operation.

Card 2: Application Software vs. OS

Q: How does application software differ from an OS?

A: Application software is designed for specific tasks, like document creation (e.g., Microsoft Word), while an OS manages hardware resources and provides a platform for running such applications.

Card 3: OS and Hardware Dependence

Q: Why is an OS hardware-dependent?

A: It is designed for specific hardware architectures, e.g., Windows works with x86/x64, macOS is built for Apple hardware, and Android is optimized for ARM architecture.

Card 4: Components of an OS

Q: Name the main components of an OS.

A: Kernel, File System, Process Management, Memory Management, Device Management, User Interface (UI).

Card 5: Kernel's Role

Q: What is the kernel in an OS?

A: The core of the OS that directly interacts with hardware, managing resources like CPU, memory, and devices.

Card 6: Examples of File Systems

Q: Give examples of file systems.

A: NTFS in Windows, ext4 in Linux.

Card 7: Basic Computer Components for OS

Q: What are the key components of a computer related to OS?

A: CPU, Memory (RAM), Input/Output devices, and Storage Devices.

Card 8: Types of Operating Systems

Q: What are the types of Operating Systems?

A: Desktop OS, Mobile OS, Embedded System OS, Real-Time OS (RTOS), Server OS, Hybrid OS.

Card 9: Functions of an OS

Q: List primary functions of an OS.

A: Process Management, Memory Management, File Management, Device Management, Security, Networking, Error Detection.

Card 10: User vs. Kernel Space

Q: What is the difference between User Space and Kernel Space?

A: User Space is where applications run with limited hardware access, while Kernel Space handles core OS functions with full hardware access.

Card 11: Modes of Operation

Q: What are the OS modes?

A: User Mode (restricted access) and Kernel Mode (full hardware access).

Card 12: Interrupts

Q: What is an interrupt?

A: A signal alerting the CPU to immediate tasks, e.g., a keyboard press.

Card 13: System Calls

Q: What are system calls?

A: Requests made by user-space applications for OS services, e.g., reading a file.

Would you like to add, modify, or reorganize any of these cards?